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Method for switching pairing in mesh network system

Abstract

Disclosed is a method for switching pairing in a mesh network system, including: selecting, by a network node, a new parent node; generating and sending, by the network node, a pairing change request packet to the new parent node when the new parent node is not an access point; adding, by the new parent node, MAC addresses of the network node and its descendant nodes to a refugee table when determining that it can provide relay services; returning, by the new parent node, a first pairing approval packet to the network node when the new parent node is a root node; forwarding, by the new parent node, the pairing change request packet upstream when the new parent node is not the root node; wherein pairing switching is successful if the network node receives the first pairing approval packet.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
(1) This application claims the priority benefit of Chinese Patent Application Serial Number 202210795773.8, filed on Jul. 7, 2022, the full disclosure of which is incorporated herein by reference.
BACKGROUND
Technical Field
(2) The present disclosure relates to the field of communication technologies, and in particular, to a method for switching pairing in a mesh network system.
Related Art
(3) In recent years, with the rapid development of network technology, the mesh network has been widely used in the fields of Internet of Things (IoT), routers and the like.
(4) In an existing mesh network, when a node takes switching action, it is disconnected with the current parent node first, then selects a new parent node to connect, and then updates a routing table of itself due to changes in the network topology of the existing mesh network.
(5) However, there are many different networking technologies for mesh networks. Different networking technologies may have different methods for switching pairing, so that relevant nodes in the network can obtain and update node switching information at the same time to meet the data transmission requirements of the mesh network.
SUMMARY
(6) Embodiments of the present disclosure provide a method for switching pairing in a mesh network system, which can allow a network node to quickly and efficiently switch to pairing with a new parent node, and allow the relevant nodes in a mesh network to obtain and update switching information of the network node at the same time.
(7) In order to solve above-mentioned technical problem, the present disclosure is implemented as follows.
(8) The present disclosure provides a method for switching pairing in a mesh network system, comprising: performing, by a network node, a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node; generating and sending, by the network node, a pairing change request packet carrying a MAC address of its descendant node, a MAC address of an old root node, and a MAC address of the old parent node to the new parent node when the new parent node is not an access point, wherein a sending address of the pairing change request packet is a MAC address of the network node and a receiving address of the pairing change request packet is a MAC address of the new parent node, the old root node is a root node corresponding to the old parent node, and the pairing change request packet selectively carries the MAC address of the new parent node; adding, by the new parent node receiving the pairing change

request packet, MAC addresses of the network node and its descendant node to a refugee table stored by itself when determining that it can provide relay services for the network node and its descendant node, wherein the refugee table stored by the new parent node contains MAC addresses of all subordinate nodes that the new parent node provides relay services for.
(9) The present disclosure provides another method for switching pairing in a mesh network system, comprising: performing, by a network node, a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node; generating and sending, by the network node, a pairing switching packet carrying a pairing switch identifier, MAC addresses of the network node and its descendant node, and a MAC address of the old parent node to an access point when the new parent node is the access point, so that the access point forwards the pairing switching packet to an old root node, wherein a receiving address of the pairing switching packet is a MAC address of the access point, a sending address of the pairing switching packet is the MAC address of the network node, and a destination address of the pairing switching packet is a MAC address of the old root node; deleting, by the old root node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the pairing switching packet, and generating and sending a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, a sending address of the second pairing cancellation packet is the MAC address of the old root node, and the refugee table stored by the old root node contains MAC addresses of all subordinate nodes that the old root node provides relay services for; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.
(10) The present disclosure provides yet another method for switching pairing in a mesh network system, comprising: performing, by a network node, a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node; generating and sending, by the network node, a pairing change request packet carrying a MAC address of a descendant node of the network node, a MAC address of an old root node, and a MAC address of the old parent node to the new parent node when the new parent node is not an access point, wherein a sending address of the pairing change request packet is a MAC address of the network node, a receiving address of the pairing change request packet is a MAC address of the new parent node, and the old root node is a root node corresponding to the old parent node; generating and returning, by the new parent node receiving the pairing change request packet, a first pairing approval packet to the network node after determining that the MAC addresses of the network node and its descendant node are contained in a refugee table stored by itself, and generating and sending a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein the refugee table stored by the new parent node contains MAC addresses of all subordinate nodes that the new parent node provides relay services for, a sending address of the first pairing approval packet is the MAC address of the new parent node, a receiving address of the first pairing approval packet is the MAC address of the network node, a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the new parent node; wherein pairing switching is successful if the network node receives the first pairing approval packet; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant nodes from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself,

and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.
(11) In the embodiments of the present disclosure, the method for switching pairing in the mesh network system can be applied to a mesh network that carries out packet forwarding based on the MAC address, and the update of the refugee tables of the relevant nodes in the mesh network is completed during the pairing switching process, which can achieve fast, stable and efficient pairing switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The features of the exemplary embodiments believed to be novel and the elements and/or the steps characteristic of the exemplary embodiments are set forth with particularity in the appended claims. The Figures are for illustration purposes only and are not drawn to scale. The exemplary embodiments, both as to organization and method of operation, may best be understood by reference to the detailed description which follows taken in conjunction with the accompanying drawings in which:
(2) FIG. **1** is a flow chart of a method for pairing nodes according to a first embodiment of the present disclosure;
(3) FIG. **2** is a schematic diagram of a format of a pairing request packet according to an embodiment of the present disclosure;
(4) FIG. **3** is a schematic diagram of a format of a pairing approval packet according to an embodiment of the present disclosure;
(5) FIG. **4** is the schematic diagram of a mesh network system according to a first embodiment of the present disclosure;
(6) FIG. **5** is a schematic diagram of a relay node according to an embodiment of the present disclosure;
(7) FIG. **6** is a schematic diagram of a node to be paired according to an embodiment of the present disclosure;
(8) FIG. **7** and FIG. **8** are a flowchart of a first embodiment of a method for pairing nodes in which the node to be paired in FIG. **6** joins the mesh network system in FIG. **4**;
(9) FIG. **9** is a flowchart of a method for pairing nodes according to a second embodiment of the present disclosure;
(10) FIG. **10** is a schematic diagram of a format of a pairing rejection packet according to an embodiment of the present disclosure;
(11) FIG. **11** is a flowchart of a second embodiment of a method for pairing nodes in which the node to be paired in FIG. **6** joins the mesh network system in FIG. **4**;
(12) FIG. **12** is a flowchart of a method for pairing nodes according to a third embodiment of the present disclosure;
(13) FIG. **13** is a flowchart of a method for pairing nodes according to a fourth embodiment of the present disclosure;
(14) FIG. **14** is a flowchart of a data transmission method according to a first embodiment of the present disclosure;
(15) FIG. **15** is a schematic diagram of a format of a packet generated by a station according to an embodiment the present disclosure;
(16) FIG. **16** is a flowchart of a data transmission method according to a second embodiment of the present disclosure;

(17) FIG. **17** and FIG. **18** are a flowchart of a method for switching pairing in a mesh network system according to a first embodiment of the present disclosure;
(18) FIG. **19** and FIG. **20** are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **17** and FIG. **18**;
(19) FIG. **21** is a flow chart of a method for switching pairing in a mesh network system according to a second embodiment of the present disclosure;
(20) FIG. **22** and FIG. **23** are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **21**;
(21) FIG. **24** is a flow chart of a method for switching pairing in a mesh network system according to a third embodiment of the present disclosure;
(22) FIG. **25** is a flow chart of a method for switching pairing in a mesh network system according to a fourth embodiment of the present disclosure;
(23) FIG. **26** and FIG. **27** are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **25**;
(24) FIG. **28** is a flow chart of a method for switching pairing in a mesh network system according to a fifth embodiment of the present disclosure;
(25) FIG. **29** and FIG. **30** are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **28**;
(26) FIG. **31** is a flow chart of a method for switching pairing in a mesh network system according to a sixth embodiment of the present disclosure;
(27) FIG. **32** and FIG. **33** are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **31**; and
(28) FIG. **34** is a flow chart of a method for switching pairing in a mesh network system according to a seventh embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(29) The following embodiments describe the features and advantages of the present disclosure in detail, but do not limit the scope of the present disclosure in any point of view. According to the description, claims, and drawings, a person ordinarily skilled in the art can easily understand the technical content of the present disclosure and implement it accordingly.
(30) The embodiments of the present disclosure will be described below in conjunction with the relevant drawings. In the figures, the same reference numbers refer to the same or similar components or method flows.
(31) It must be understood that the words “including”, “comprising” and the like used in this specification are used to indicate the existence of specific technical features, values, method steps, work processes, elements and/or components. However, it does not exclude that more technical features, values, method steps, work processes, elements, components, or any combination of the above can be added.
(32) It must be understood that when an element is described as being “connected” or “coupled” to another element, it may be directly connected or coupled to another element, and intermediate elements therebetween may be present. In contrast, when an element is described as being “directly connected” or “directly coupled” to another element, there is no intervening element therebetween.
(33) Before a method for switching pairing in a mesh network system of the present disclosure is described, the nouns of the present disclosure are explained first. The noun “mesh network system” mentioned in the present disclosure refers to a mesh communication network formed by network nodes communicating with each other. The noun “network node” refers to a computing device connected to a network with independent addressing, transmitting or receiving data capabilities, such as a workstation, a server, a terminal device, and a network device, wherein the network node with the relay function is called a relay node, and the network node without the relay function is called a station (STA). All network nodes can provide the STA function working as a station in the infrastructure mode defined by the 802.11 protocol, so the relay node can execute different

programs to play the role of a station. It should be noted that, for a relay node, the MAC addresses used for the STA function and the MAC addresses used for the relay function can be the same or different.

(34) The network node/relay node directly connected to an access point (AP) is called a root node. The network node connected to the access point through the root node is called a second-level node, the network node connected to the access point through the second-level node is called a third-level node, and so on. For two network nodes that are directly connected to each other, the upper-level network node is called a parent node of the lower-level network node, and the lower-level network node is called a child node of the upper-level network node. Each network node has only one parent node, and a parent node can have many child nodes. The root node is directly connected to the access point, and the access point can be considered as the parent node of the root node. The network node whose level is greater than the level of a certain network node is called the subordinate node of the certain network node. For example, the third-level node, the fourth-level node, and the fifth-level node are all subordinate nodes of the second-level node. All subordinate nodes that need to communicate with the access point through a certain network node are called descendant nodes of the certain network node.

(35) In addition, the “refugee table” stored in a relay node mentioned in the present disclosure is used for recording the relevant information of the descendant nodes of the relay node, such as a MAC address and a variant on the MAC address, wherein the variant on the MAC address is the result of the hash calculation of the MAC address, or the result of the cyclic redundancy check (CRC) calculation of the MAC address. In the following embodiments, the refugee table stored in a relay node is used for recording the MAC addresses of descendant nodes of the relay node for illustration. Moreover, the “node to be paired” mentioned in the present disclosure refers to a new node that intends to join the mesh network system, which may or may not have the relay capability; when the node to be paired has the relay capability, it can have the same module and structure as the relay node, and play different roles because it executes different programs. Furthermore, the “MAC address of the access point” mentioned in the present disclosure refers to the basic service set identifier (BSSID) of the access point.

(36) During the construction process of the mesh network system, each relay node can establish the content of the refugee table stored by itself, and each network node can confirm its own parent node. Therefore, it is necessary to explain the construction process of the mesh network system before a method for switching pairing in a mesh network system of the present disclosure is described.

(37) In the construction process of the mesh network system, the network node as the root node can establish its connection with the access point through the existing method for pairing nodes, which is not described herein. The network node as the non-root node (hereinafter referred to as a node to be paired) can confirm its own parent node and join the mesh network system through the following method for pairing nodes, and make each relay node of the mesh network system update the content of the refugee table stored by itself.

(38) Please refer to FIG. **1**, which is a flow chart of a method for pairing nodes according to a first embodiment of the present disclosure. The method for pairing nodes comprises: sending, by a node to be paired, a pairing request packet to its parent node, wherein a sending address of the pairing request packet is a MAC address of the node to be paired, a receiving address of the pairing request packet is a MAC address of the parent node of the node to be paired, and the parent node of the node to be paired is a relay node of a mesh network system (step **110**); adding, by a relay node, the MAC address of the node to be paired to an refugee table stored by itself after the relay node receiving the pairing request packet determines that it can provide a relay service for the node to be paired, wherein the refugee table contains MAC addresses of all subordinate nodes that the relay node provide relay services for (step **120**); changing, by the relay node, the receiving address of the pairing request packet received to a MAC address of its parent node if the relay node is a non-root

node, and then forwarding the pairing request packet upstream until the relay node receiving the pairing request packet is a root node (step **130**); returning, by the relay node, a pairing approval packet if the relay node is the root node, wherein a receiving address of the pairing approval packet is the MAC address of the node to be paired, and a sending address of the pairing approval packet is a MAC address of itself (step **140**); changing, by a relay node receiving the pairing approval packet, the sending address of the pairing approval packet to a MAC address of itself when the relay node receiving the pairing approval packet confirms that the sending address of the pairing approval packet is a MAC address of its parent node, and a receiving address is contained in the refugee table stored by itself, and then forwarding the pairing approval packet downstream until the pairing approval packet is forwarded to the node to be paired (step **150**); a pairing being successful if the node to be paired receives the pairing approval packet (step **160**). It should be noted that, when the parent node of the node to be paired is the root node, step **130** may be omitted.
(39) In an embodiment, when the parent node of the node to be paired is a non-root node, after step **140** and before step **160**, the method for pairing nodes may further comprise: changing, by the relay node receiving the pairing approval packet, the sending address of the pairing approval packet to a MAC address of itself when it confirms that the sending address of the pairing approval packet is a MAC address of its parent node and the receiving address is contained in the refugee table stored by itself, and then forwarding the pairing approval packet downstream until the pairing request packet is forwarded to the node to be paired (step **150**).
(40) Each of the format of the pairing request packet and the pairing approval packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames, and the pairing request packet and the pairing approval packet can use the same type of packet having different meanings or use different types of packets.
(41) In one embodiment, the format of the pairing request packet may be a null data frame defined by the 802.11 protocol. The null data frame comprises a frame control field, a duration field, an address 1 field, an address 2 field, an address 3 field, a sequence control field, and a frame check sequence field. Specifically, sending, by the node to be paired, the pairing request packet in step **110** comprises: filling, by the node to be paired, an address 1 field in a null data frame with the MAC address of the parent node of the node to be paired, filling an address 2 field in the null data frame with the MAC address of the node to be paired, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a third preset value to define a type of the handshake frame as a pairing request, so as to generate and send the pairing request packet. In an example, the first preset bit of the address 3 field may be bit**0**, the first preset value may be 1 (that is, bit**0** is 1), the plurality of second preset bits of the address 3 field may be bit**1** to bit**4**, the third preset value may be 0 (that is, bit**1** to bit**4** are all 0), and other bits of the address 3 field (that is, bit**5** to bit**47**) may be reserved for encoding, so that the handshake frame carries the pairing related information, as shown in FIG. **2**, which is a schematic diagram of a format of a pairing request packet according to an embodiment of the present disclosure. It should be noted that although the value of bit**0** of the address 3 field defined by the 802.11 protocol is 1, which indicates a multicast address, the null data frame does not include a multicast address in actual use, so that this feature can be used to allow hardware and software to easily recognize that it is the packet defined in the present disclosure.
(42) In addition, the format of the pairing approval packet can also be a null data frame defined by the 802.11 protocol. Specifically, returning, by the relay node, a pairing approval packet if it is the root node in step **140**, comprises: filling, by the root node, an address 1 field in a null data frame with the MAC address of the node to be paired, filling an address 2 field in the null data frame with a MAC address of the root node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a fourth preset value to define a

type of the handshake frame as pairing approval, so as to generate and return the pairing approval packet. In an example, the first preset bit of the address 3 field may be bit**0**, the first preset value may be 1 (that is, bit**0** is 1), the plurality of second preset bits of the address 3 field may be bit**1** to bit**4**, the fourth preset value may be 1 (that is, bit**1** to bit**3** are all 0, and bit**4** is 1), and other bits of the address 3 field (that is, bit**5** to bit**47**) may be reserved for encoding, so that the handshake frame carries the pairing related information as shown in FIG. **3**, which is a schematic diagram of a format of a pairing approval packet according to an embodiment of the present disclosure.
(43) More specifically, the description of the method for pairing nodes shown in FIG. **1** is described below in conjunction with FIG. **4** to FIG. **8** as an embodiment.
(44) Please refer to FIG. **4**, which is a schematic diagram of a mesh network system according to an embodiment of the present disclosure. As shown in FIG. **4**, the mesh network system **100** comprises an access point **200** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprise a plurality of relay nodes (i.e., the relay node **300***a*, the relay node **300***b*, and the relay node **300***c*), wherein the relay node **300***a* serves as the root node communicating with the access point **200**, the parent node of the relay node **300***c* is the relay node **300***b*, and the parent node of the relay node **300***b* is the relay node **300***a*.
(45) Please refer to FIG. **5**, which is a schematic diagram of a relay node according to an embodiment of the present disclosure. The relay node **300***a*, the relay node **300***b* and the relay node **300***c* may all comprise: a storage module **410**, a receiving module **420**, a processing module **430**, a packet generating module **440**, an address change module **450** and a sending module **460**, wherein the receiving module **420** is connected to the storage module **410**, the processing module **430** is connected to the receiving module **420** and the storage module **410**, the packet generation module **440** is connected to the processing module **430**, the address change module **450** is connected to the storage module **410** and the processing module **430**, the sending module **460** is connected to the address change module **450** and the packet generation module **440**, and the storage module **410** is configured to store the refugee table. The receiving module **420**, the processing module **430**, the packet generating module **440**, the address change module **450** and the sending module **460** may be implemented as software, hardware, and an appropriate combination thereof. The software module may be stored in a storage medium in the art, and do its work through the processor. For example, if the above-mentioned modules are all software modules, they can work by the same processor, or by different processors in any combination.
(46) Please refer to FIG. **6**, which is a schematic diagram of a node to be paired according to an embodiment of the present disclosure. The node to be paired **500** is a node that intends to join the mesh network system **100**. The node to be paired **500** comprise a packet generation module **510**, a sending module **520**, a receiving module **530**, and a processing module **540**, wherein the sending module **520** is connected to the packet generation module **510**, and the processing module **540** is connected to the sending module **520** and the receiving module **530**, and the processing module **540** comprises a timer **542**. The packet generation module **510**, the sending module **520**, the receiving module **530** and the processing module **540** can be implemented as software, hardware, and an appropriate combination thereof. The software module may be stored in a storage medium in the art, and do its work through the processor. For example, if the above-mentioned modules are all software modules, they can work by the same processor, or by different processors in any combination.
(47) Please refer to FIG. **4** to FIG. **8**, wherein FIG. **7** and FIG. **8** are a flowchart of a first embodiment of a method for pairing nodes in which the node to be paired in FIG. **6** joins the mesh network system in FIG. **4**. When the to-be-paired node **500** in FIG. **6** intends to join the mesh network system **100** in FIG. **4**, the to-be-paired node **500** generates a pairing request packet by the packet generation module **510** thereof (step **610**), the to-be-paired node **500** sends the pairing request packet to the relay node **300***c* by the sending module **520** thereof (step **620**), and starts the

timer **542** of the to-be-paired node **500** (step **630**), wherein the parent node of the node to be paired **500** is the relay node **300***c*, the sending address of the pairing request packet is the MAC address of the node to be paired **500**, and the receiving address of the pairing request packet is the MAC address of the relay node **300***c*. Since the receiving address of the pairing request packet is the MAC address of the relay node **300***c*, the relay node **300***c* receives the pairing request packet by the receiving module **420**. When the processing module **430** of the relay node **300***c* determines that it can provide the relay service for the node to be paired **500**, based on the sending address of the pairing request packet being the MAC address of the node to be paired **500**, the MAC address of the node to be paired **500** is added to the refugee table stored in storage module **410** of the relay node **300***c* (step **640**), and the relay node **300***c* determines whether it is the root node. When the processing module **430** of the relay node **300***c* determines that the relay node **300***c* is a non-root node, the relay node **300***c* changes the receiving address of the received pairing request packet from the MAC address of the relay node **300***c* to the MAC address of the relay node **300***b* by the address change module **450** thereof (step **650**), and the sending module **460** of the relay node **300***c* sends the pairing request packet with the receiving address changed by the address change module **450** to the relay node **300***b* (step **660**).

(48) Since the receiving address of the pairing request packet is the MAC address of the relay node **300***b*, the relay node **300***b* receives the pairing request packet by the receiving module **420** thereof. When the processing module **430** of the relay node **300***b* determines that the relay node **300***b* can provide the relay service for the node to be paired **500**, based on the sending address of the pairing request packet being the MAC address of the node to be paired **500**, the MAC address of the node to be paired **500** is added to the refugee table stored in the storage module **410** of the relay node **300***b* (step **670**). When the processing module **430** of the relay node **300***b* determines that the relay node **300***b* is a non-root node, the relay node **300***b* changes the receiving address of the received pairing request packet from the MAC address of the relay node **300***b* to the MAC address of the relay node **300***a* by the address change module **450** thereof (step **680**), and the sending module **460** of the relay node **300***b* sends the pairing request packet whose receiving address is changed by the address change module **450** to the relay node **300***a* (step **690**).

(49) Since the receiving address of the pairing request packet is the MAC address of the relay node **300***a*, the relay node **300***a* receives the pairing request packet by the receiving module **420** thereof. When the processing module **430** of the relay node **300***a* determines that the relay node **300***a* can provide the relay service for the node to be paired **500**, based on the sending address of the pairing request packet being the MAC address of the node to be paired **500**, the MAC address of the node to be paired **500** is added to the refugee table stored in the storage module **410** of the relay node **300***a* (step **710**). When the processing module **430** of the relay node **300***a* determines that the relay node **300***a* is the root node, the relay node **300***a* generates a pairing approval packet by the packet generation module **440** thereof (step **720**), and the sending module **460** of the relay node **300***a* sends the pairing approval packet generated by the packet generation module **440** to the relay node **300***b* (step **730**), wherein the receiving address of the pairing approval packet is the MAC address of the node to be paired **500** and the sending address of the pairing approval packet is the MAC address of the relay node **300***a*.

(50) Since the sending address of the pairing approval packet is the MAC address of the relay node **300***a*, and the receiving address of the pairing approval packet is contained in the refugee table stored in the storage module **410** of the relay node **300***b*, the relay node **300***b* receives the pairing approval packet by the receiving module **420** thereof, and changes the sending address of the pairing approval packet to the MAC address of the relay node **300***b* by the address change module **450** thereof (step **740**), and the sending module **460** of the relay node **300***b* sends the pairing approval packet whose sending address is changed by the address change module **450** to the relay node **300***c* (step **750**). Since the sending address of the pairing approval packet is the MAC address of the relay node **300***b*, and the receiving address is contained in the refugee table stored in the

storage module **410** of the relay node **300***c*, the relay node **300***c* receives the pairing approval packet by receiving module **420** thereof, and changes the sending address of the pairing approval packet to the MAC address of the relay node **300***c* by the address change module **450** thereof (step **760**), and the sending module **460** of the relay node **300***c* sends the pairing approval packet whose sending address is changed by the address change module **450** to the node to be paired **500** (step **770**).
(51) Since the receiving address of the pairing approval packet is the MAC address of the node to be paired **500**, the node to be paired **500** receives the pairing approval packet by the receiving module **530** thereof. If the receiving module **530** of the node to be paired **500** receives the pairing approval packet, and the timer **542** of the to-be-paired node **500** does not expire, it means that the pairing is successful (step **780**) (that is, the node to be paired **500** successfully joins the mesh network system **100**).
(52) Please refer to FIG. **9**, which is a flowchart of a method for pairing nodes according to a second embodiment of the present disclosure. In addition to the step **110** to step **160** described above, the method for pairing nodes may further comprise: returning, by the relay node receiving the pairing request packet, a pairing rejection packet when determining that it cannot provide the relay service for the node to be paired, wherein a receiving address of the pairing rejection packet is the MAC address of the node to be paired and a sending address of the pairing rejection packet is a MAC address of itself (step **170**); and the pairing failing if the node to be paired receives the pairing rejection packet (step **190**). It should be noted that, when the relay node determining that it cannot provide the relay service for the node to be paired and returning the pairing rejection packet is not the parent node of the node to be paired, after step **170** and before step **190**, the method for pairing nodes may further comprise: deleting, by the relay node receiving the pairing rejection packet, the MAC address of the node to be paired from a refugee table stored by itself when determining that the sending address of the pairing rejection packet is a MAC address of its parent node and the receiving address of the pairing rejection packet is contained in the refugee table stored by itself, and changing the sending address of the pairing rejection packet received to a MAC address of itself, and then forwarding the pairing rejection packet downstream until the pairing rejection packet is forwarded to the node to be paired (step **180**).
(53) The format of the pairing rejection packet may be a data frame defined by the 802.11 protocol, such as a management frame, a control frame, and other self-defined frames. In an embodiment, the format of the pairing rejection packet may also be a null data frame defined by the 802.11 protocol. Specifically, the step **170** of returning, by the relay node receiving the pairing request packet, a pairing rejection packet when determining that it cannot provide the relay service for the node to be paired further comprises: filling, by the relay node receiving the pairing request packet, an address 1 field in a null data frame with the MAC address of the node to be paired, filling an address 2 field in the null data frame with a MAC address of itself, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a second preset value to define a type of the handshake frame as a pairing rejection, so as to generate and return the pairing rejection packet. In an example, the first preset bit of the address 3 field may be bit**0**, the first preset value may be 1, the plurality of second preset bits of the address 3 field may be bit**1** to bit**4**, the second preset value may be 2 (that is, bit**1**, bit**2**, and bit**4** are all 0, and bit**3** is 1), and the other bits of the address 3 field (that is, bit**5** to bit**47**) may be reserved for encoding, so that the handshake frame carries the pairing related information, as shown in FIG. **10**, which is a schematic diagram of a format of a pairing rejection packet according to an embodiment of the present disclosure.
(54) In an embodiment, the situation that the relay node determines that it cannot provide the relay service for the node to be paired may comprise but not limited to: (1) the situation where the storage capacity of the refugee table of the relay node has reached a upper limit value, so that the

relay node can no longer provide relay services for more network nodes; (2) the situation where the relay node shuts down the relay service function; (3) the situation where the relay node is trying to switch to connect to other network nodes; and (4) the situation where the relay node has left the mesh network system.
(55) It should be noted that, the relay node determining that it cannot provide the relay service for the node to be paired in step **170** may be a root node or a non-root node.
(56) More specifically, the description of the method for pairing nodes shown in FIG. **9** is described below in conjunction with FIG. **4** to FIG. **6** and FIG. **11** as an embodiment.
(57) Please refer to FIG. **4** to FIG. **6** and FIG. **11**, wherein FIG. **11** is a flowchart of a second embodiment of a method for pairing nodes in which the node to be paired in FIG. **6** joins the mesh network system in FIG. **4**. Since the difference between FIG. **11** and FIGS. **7-8** lies in the process after the relay node **300***b* determines that it cannot provide the relay service for the node to be paired **500**, the description of the process from step **610** to step **660** will not be repeated. As shown in FIG. **11**, after the relay node **300***b* determines that it cannot provide the relay service for the node to be paired **500** by the processing module **430** thereof, the relay node **300***b* generates a pairing rejection packet by the packet generation module **440** thereof (step **790**), and sends the pairing rejection packet to the relay node **300***c* by the sending module **460** thereof (step **810**), wherein the receiving address of the pairing rejection packet is the MAC address of the node to be paired **500** and the sending address of the pairing rejection packet is the MAC address of the relay node **300***b*. After the relay node **300***c* confirms that the sending address of the pairing rejection packet is the MAC address of its parent node by the processing module **430** thereof, and the receiving address of the pairing rejection packet is contained in the refugee table stored in the storage module **410** of the relay node **300***c*, the relay node **300***c* deletes the MAC address of the node to be paired **500** from the refugee table stored in the storage module **410** of the relay node **300***c* (step **820**). Then, the relay node **300***c* changes the sending address of the received pairing rejection packet to the MAC address of the relay node **300***c* by the address change module **450** thereof (step **830**). After that, the relay node **300***c* sends the pairing rejection packet whose sending address is changed by the address change module **450** to the node to be paired **500** by the sending module **460** thereof (step **840**).
(58) Since the receiving address of the pairing rejection packet is the MAC address of the node to be paired **500**, the node to be paired **500** receives the pairing rejection packet by the receiving module **530** thereof. If the receiving module **530** of the node to be paired **500** receives the pairing rejection packet, and the timer **542** of the node to be paired **500** does not expire, it means that the pairing fails (step **850**) (that is, the node to be paired **500** fails to join the mesh network system **100**).
(59) Please refer to FIG. **12**, which is a flowchart of a method for pairing nodes according to a third embodiment of the present disclosure. In addition to step **110** to step **190**, the method for pairing nodes may further comprise: the pairing failing if the timer expires (step **192**). Specifically, if the timer expires and the node to be paired does not receive the pairing rejection packet or the pairing approval packet, it means that the pairing fails.
(60) In one embodiment, please refer to FIG. **13**, which is a flowchart of a method for pairing nodes according to a fourth embodiment of the present disclosure. Before step **110**, the method for pairing nodes may further comprise: setting the node to be paired to only receive various types of packets that meet a filtering condition, wherein the filtering condition is that a sending address of each type of packets is the MAC address of the parent node of the node to be paired and a receiving address of each type of packets is the MAC address of the node to be paired (step **102**).
(61) In an embodiment, please refer to FIG. **13**, before step **102**, the method for pairing nodes may further comprise: obtaining, by the node to be paired, a received signal strength indication (RSSI) value between each relay node and an access point in the mesh network system and a RSSI value between the node to be paired and each relay node based on multiple beacon packets or multiple probe response packets obtained by a scanning procedure (step **104**); and selecting, by the node to

be paired, a relay node as its parent node based on the RSSI value between each relay node and the access point and the RSSI value between the node to be paired and each relay node (step **106**).
(62) In an example, the scanning procedure is that the node to be paired actively detects and searches each relay node of the mesh network system. Specifically, the node to be paired may send probe request frames sequentially on the channels it supports, wherein the probe request frame does not carry the SSID but may comprise vendor-defined information elements (Vendor IEs), such as the BSSID and the information requesting assistance, to detect all available relay nodes around. The relay node receiving the probe request frame sends a probe response frame to the node to be paired, wherein the probe response frame may comprise Vendor IEs, such as the BSSID, the RSSI value between itself and the access point, the MAC address of the root node it is connected to, the number of levels it is at in the mesh network system, the number of descendant nodes that provided with relay services by itself, and information that it can provide assistance. Therefore, the node to be paired can receive the probe response frame sent by each relay node through the above scanning procedure, and obtain the RSSI value between itself and each relay node and the RSSI value between each relay node and the access point, and then select a relay node as its parent node. For example, the node to be paired first selects the relay node with the best RSSI value as its parent node, but if the RSSI value between the selected relay node and the parent node thereof is poor or the RSSI value between the nodes on the uplink path of the selected relay node is poor, the node to be paired chooses the relay node with the second best RSSI value as its parent node, and so on, until the node to be paired selects a suitable relay node as its parent node.
(63) In another example, the scanning procedure is that the node to be paired passively receives the beacon frame sent by each relay node of the mesh network system. Specifically, each relay node of the mesh network system periodically broadcasts the beacon frame, and the node to be paired can listen to the beacon frames on each channel it supports, wherein the beacon frame can comprise Vendor IEs, such as the BSSID, the RSSI value between itself and the access point, the MAC address of the root node it is connected to, the number of levels it is at in the mesh network system, the number of descendant nodes that provided with relay services by itself, and information that it can provide assistance. Therefore, the node to be paired can receive the probe response frame sent by each relay node through the above scanning procedure, and obtain the RSSI value between the node to be paired and each relay node and the RSSI value between each relay node and the access point, and then select a relay node as its parent node.
(64) Through the methods for pairing nodes described in FIG. **1**, FIG. **9**, FIG. **12** and FIG. **13**, the network node as the non-root node (that is, the node to be paired) can confirm its parent node and join the mesh network system, and each relay node of the mesh network system can update the content of the refugee table stored by itself, so as to construct the mesh network system, and apply to the following data transmission method and the method for switching pairing of the present disclosure.
(65) Please refer to FIG. **14**, which is a flowchart of a data transmission method according to a first embodiment of the present disclosure. The data transmission method comprises: generating a packet by a station, wherein a sending address of the packet is a MAC address of the station and a receiving address of the packet is a MAC address of an access point (step **1010**); changing the receiving address of the packet to a MAC address of its parent node by the station, and then sending the packet (step **1020**); transmitting an acknowledgment (ACK) frame by a relay node that receives the packet, changing the receiving address of the packet to a MAC address of its parent node, and then forwarding the packet upstream until the packet is forwarded to the access point (step **1030**); and transmitting an ACK frame by the access point after receiving the packet, parsing the packet, and determining that it receives data from the station based on the sending address of the packet (step **1040**).
(66) Generally, network integrated circuits (ICs) mainly process data in a physical layer (i.e., Layer 1) and a data link layer (i.e., Layer 2) with hardware, and process data in above Layer 3 with

software. However, with the increasing demand for bandwidth, the requirements for the transmission rate of routers have also increased significantly. Therefore, it is also possible to choose to process the data of the third layer by hardware for the network ICs. In this embodiment, the station can include a software layer and a hardware layer, so in step **1010**, the station can generate a packet by the software layer, and in step **1020**, the station can change the receiving address of the packet to the MAC address of its parent node by the hardware layer, and then send the packet. In another embodiment, the station can change the receiving address of the packet to the MAC address of its parent node by the software layer, and then send the packet.
(67) In one embodiment, the format of the packet generated by the software layer of the station may use a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames.
(68) In one embodiment, the format of the packet generated by the software layer of the station may be a data frame defined by the 802.11 protocol, and the data frame comprises a frame control field, a duration field, an address 1 field, an address 2 field, an address 3 field, a sequence control field, an address 4 field, a quality of service (QoS) control field, a high throughput (HT) control field, a frame body field, and a frame check sequence (FCS) field, as shown in FIG. **15**, which is a schematic diagram of a format of a packet generated by a station according to an embodiment of the present disclosure. Specifically, the software layer of the station fills the address 1 field in the data frame with the MAC address of the access point, and fills the address 2 field with the MAC address of itself to generate the packet, but this embodiment is not intended to limit the present disclosure.
(69) In one embodiment, in order to consider the security of data transmission, the data transmission method may be further performed by encrypting the packet at the station, and decrypting the packet at the access point. Specifically, when the encryption and decryption method is performed by using the Temporal Key Integrity Protocol (TKIP), the Counter Mode Cipher Block Chaining Message Authentication Protocol (CCMP) or the Wired Equivalent Privacy Protocol (WEP), step **1010** may comprise: generating the packet by the software layer of the station, and then encrypting the packet, or step **1020** may comprise: changing, by the hardware layer of the station, the receiving address of the packet to the MAC address of its parent node after the hardware layer of the station encrypts the packet, and then sending the packet; step **1040** may comprise: decrypting, by the access point, the packet received, and then determining that it receives the data from the station based on the sending address of the packet being the MAC address of the station. In an example, the station may encrypt the packet by the software layer or the hardware layer, then change the receiving address of the packet to the MAC address of its parent node, and then send the packet.
(70) In another embodiment, when the encryption and decryption method is performed by using the TKIP or WEP, step **1020** may comprise: encrypting, by the station, the packet after the hardware layer of the station changes the receiving address of the packet to the MAC address of its parent node, and then sending the packet; step **1040** may comprise: decrypting, by the access point, the packet received, and then determining that it receive the data from the station based on the sending address of the packet being the MAC address of the station.
(71) In one embodiment, in order to ensure the correctness and integrity of data transmission, the data transmission method may use a cyclic redundancy check (CRC) algorithm to check the packet. Specifically, step **1020** may comprise: performing, by the station, CRC calculation on the packet, and fill a calculated CRC code into the packet after the hardware layer of the station changes the receiving address of the packet to the MAC address of its parent node, and then sending the packet; step **1030** may comprise: performing, by the relay node receiving the packet, CRC check on the packet, returning an ACK frame when the packet passes the CRC check, changing the receiving address of the packet received to the MAC address of its parent node, performing CRC calculation on the packet, filling a calculated CRC code into the packet, and then forwarding the packet

upstream until the packet is forwarded to the access point; step **1040** may comprise: performing, by the access point, CRC check on the packet after the access point receives the packet, returning an ACK frame when the packet passes the CRC check, and then parsing the packet to obtain the data from the station.

(72) In another embodiment where the CRC check algorithm is used to check the packet, step **1020** may comprise: performing, by the station, CRC calculation on the packet, filling a calculated CRC code into the packet, changing the receiving address of the packet to the MAC address of its parent node, and then sending the packet; step **1030** may comprise: changing, by the relay node receiving the packet, the receiving address of the packet to the MAC address of the access point, performing CRC check on the packet, returning an ACK frame when the packet passes the CRC check, changing the receiving address of the packet to the MAC address of its parent node (the packet retains the original CRC code), and then forwarding the packet upstream until the packet is forwarded to the access point; step **1040** may comprise: performing, by the access point, CRC check on the packet after the access point receives the packet, returning an ACK frame when the packet passes the CRC check, and then parsing the packet to obtain the data from the station.

(73) In one embodiment, in order to consider the security, correctness and integrity of data transmission at the same time, the data transmission method may comprise the encryption and decryption of the data field of the packet and the check of the packet by using the CRC algorithm.

(74) Please refer to FIG. **16**, which is a flowchart of a data transmission method according to a second embodiment of the present disclosure. The data transmission method comprises: sending a packet by an access point, wherein a receiving address of the packet is a MAC address of a station and a sending address of the packet is a MAC address of the access point (step **2010**); receiving the packet by a relay node based on the MAC address of the station contained in a refugee table stored by itself and the sending address of the packet being a MAC address of its parent node, transmitting an ACK frame to its parent node, and then changing the sending address of the packet to a MAC address of itself and forwarding the packet downstream until the packet is forwarded to the station, wherein the refugee table stored in the relay node contains MAC addresses of all descendant nodes that provided with relay services by the relay node (step **2020**); and receiving the packet by the station based on the sending address of the packet being the MAC address of its parent node and the receiving address of the packet being the MAC address of the station, transmitting an ACK frame, changing the sending address of the packet to the MAC address of the access point, and then determining that it receives data from the access point based on the sending address of the packet (step **2030**).

(75) In one embodiment, the format of the packet sent by the access point may be a data frame defined by the 802.11 protocol, such as a management frame, a control frame, and other self-defined frames.

(76) In one embodiment, the station may include a hardware layer and a software layer, so in step **2030**, the station may receive the packet by the hardware layer, transmit the ACK frame, and change the sending address of the received packet to the MAC address of the access point and then transmit the packet to the software layer, and the software layer of the station may determines to receive data from the access point based on the sending address of the packet.

(77) In one embodiment, the format of the packet sent by the access point may be a data frame defined by the 802.11 protocol as shown in FIG. **15**. Specifically, the access point fills the address 1 field in the data frame with the MAC address of the station, and fills the address 2 field with the MAC address of itself to generate the packet, but this embodiment is not intended to limit the present disclosure. For example, the access point fills the address 3 field in the data frame with the MAC address of the station, and fills the address 4 field with the MAC address of itself to generate the packet (that is, the address 3 field is filled with the receiving address, and the address 4 field is filled with the sending address).

(78) In one embodiment, in order to consider the security of data transmission, the data

transmission method may be further performed by encrypting the packet at the access point, and decrypting the packet at the station. Specifically, step **2010** may comprise: encrypting, by the access point, the packet, and then sending the packet; step **2030** may comprise: changing, by the station, the sending address of the packet to the MAC address of the access point after the station decrypts the packet, and then determining that it receives the data from the access point based on the sending address of the packet. In another embodiment, step **2010** may comprise: encrypting, by the access point, the packet and then sending the packet; step **2030** may comprise: changing, by t the station, the sending address of the received packet to the MAC address of the access point; and decrypting, by the station, the packet, and determining that it receive the data from the access point based on the sending address of the packet.

(79) In one embodiment, in order to ensure the correctness and integrity of data transmission, the data transmission method may use the CRC algorithm to check the packet. Specifically, step **2010** may comprise: performing, by the access point, CRC calculation on the packet, filling a calculated CRC code into the packet, and then sending the packet; step **2020** may comprise: performing, by the relay node receiving the packet, CRC check on the packet, returning an ACK frame when the packet passes the CRC check, changing the sending address of the packet to the MAC address of itself, performing CRC calculation on the packet, filling a calculated CRC code into the packet, and then forwarding the packet downstream; step **2030** may comprise: performing, by the station, CRC check on the packet, returning an ACK frame when the packet passes the CRC check, changing the sending address of the packet to the MAC address of the access point, and then determining that it receives the data from the access point based on the sending address of the packet. In an example of step **2030**, the station can perform the CRC check on the packet by the hardware layer, return the ACK frame when the packet passes the CRC check, and change the sending address of the received packet to the MAC address of the access point by the hardware layer or the software layer, and the software layer of the station determines to receive data from the access point based on the sending address of the packet.

(80) In another embodiment where the CRC check algorithm is used to check the packet, step **2010** may comprise: performing, by the access point, CRC calculation on the packet, filling a calculated CRC code into the packet, and then sends the packet; step **2020** may comprise: performing, by the relay node receiving the packet, CRC check on the packet, returning an ACK frame when the packet passes the CRC check, changing the sending address of the packet to the MAC address of itself (the packet retains the original CRC code) and then forwarding the packet downstream; step **2030** may comprise: changing, by the station, the sending address of the packet to the MAC address of the access point, performing CRC check on the packet, returning an ACK frame when the packet passes the CRC check, and then determining that it receives the data from the access point based on the sending address of the packet that passes the CRC check.

(81) In one embodiment, in order to consider the security, correctness and integrity of data transmission at the same time, the data transmission method may comprise the encryption and decryption of the data field of the packet and the check of the packet by using the CRC algorithm.

(82) Please refer to FIG. **17** and FIG. **18**, which are a flowchart of a method for switching pairing in a mesh network system according to a first embodiment of the present disclosure. As shown in FIG. **17** and FIG. **18**, the method for switching pairing in the mesh network system comprises step **201** to step **205** and step **208**.

(83) In step **201**, a network node performs a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node.

(84) In one embodiment, when the network node detects that the communication quality with the current parent node is poor or receives a first pairing cancellation packet from the current parent node, it performs the scanning procedure to select the new parent node, so that the current parent node becomes the old parent node, wherein the format of the first pairing cancellation packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control

frame, and other self-defined frames. In an example, the current parent node fills an address 1 field in a null data frame with the MAC address of the network node, fills an address 2 field in the null data frame with the MAC address of the current parent node, sets a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and sets a plurality of second preset bits of the address 3 field in the null data frame to a fifth preset value to define a type of the handshake frame as pairing cancellation, so as to generate the first pairing cancellation packet.
(85) In an example, when the network node counts that an uplink packet transmission failure rate reaches a first threshold, energy of a packet received from the current parent node is lower than a second threshold, or a beacon is continuously lost for a period of time, it represents that the network node detects that the communication quality with the current parent node is poor, so that the network node performs the scanning procedure to select the new parent node, wherein the first threshold and the second threshold can be set according to actual needs.
(86) In an example, when the current parent node detects that the communication quality with the network node is poor, receives a request from the application layer to disconnect the current network connection, or needs to connect to other networks, it sends the first pairing cancellation packet to the network node, so that the network node performs the scanning procedure to select the new parent node, but this example is not intended to limit the present disclosure.
(87) In one embodiment, the manner in which the network node performs the scanning procedure to select the new parent node may be the same as the manner in which the node to be paired performs the scanning procedure to select the parent node described in FIG. **13**, and details will not be repeated here. In another embodiment, step **201** of performing, by the network node, the scanning procedure to select the new parent node, so that the current parent node becomes the old parent node further comprises: obtaining, by the network node, communication information of each relay node in the mesh network system based on a plurality of beacon packets or a plurality of probe response packets obtained through the scanning procedure; and selecting, by the network node, a relay node as the new parent node based on the communication information of each relay node. The communication information of each relay node comprises a signal strength between each relay node and the access point, a signal strength between the network node and each relay node, a level of each relay node in the mesh network system, relay service capability of each relay node, and/or a MAC address of each relay node, wherein the relay service capability of each relay node allows the network node to determine how many subordinate nodes each relay node can provide relay services for, and the MAC address of each relay node allows the network node to confirm whether the MAC address of each relay node is contained in the refugee table stored by itself.
(88) In step **202**, when the new parent node is not an access point, the network node generates and sends a pairing change request packet carrying a MAC address of its descendant node, a MAC address of an old root node and a MAC address of the old parent node to the new parent node, and starts a timer, wherein the pairing change request packet selectively carries the MAC address of the new parent node.
(89) The old root node is a root node corresponding to the old parent node; a sending address of the pairing change request packet is the MAC address of network node, and a receiving address of the pairing change request packet is the MAC address of new parent node; and the format of the pairing change request packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames. In one embodiment, the format of the pairing change request packet can expand a payload field on the basis of a pairing request packet. Specifically, the step **202** of generating and sending, by the network node, the pairing change request packet carrying the MAC address of its descendant node, the MAC address of the old root node, and the MAC address of the old parent node to the new parent node further comprises: filling, by the network node, an address 1 field in a null data frame with the MAC address of the new parent node, filling an address 2 field in the null data frame with the MAC

address of the network node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, setting a plurality of second preset bits of the address 3 field in the null data frame to a third preset value to define a type of the handshake frame as a pairing request, and filling an payload field in the null data frame, which is expanded, with the MAC address of the descendant node of the network node, the MAC address of the old root node, and the MAC address of the old parent node, so as to generate and send the pairing change request packet to the new parent node, wherein the expanded payload field in the pairing change request packet may carry the MAC address of the new parent node.
(90) In an example, the first preset bit of the address 3 field may be bit**0**, the first preset value may be 1 (that is, bit**0** is 1), the plurality of second preset bits of the address 3 field may be bit**1** to bit**4**, the third preset value may be 0 (that is, bit**1** to bit**4** are all 0), other bits of the address 3 field (that is, bit**5** to bit**47**) may be reserved for encoding, and the expanded payload field is sequentially filled with the MAC address of the descendant node of the network node, the MAC address of the old root node, and the MAC address of the old parent node.
(91) In step **203**, when the new parent node receiving the pairing change request packet determines that it can provide relay services for the network node and its descendant node, the new parent node adds the MAC addresses of network node and its descendant node to a refugee table stored by itself.
(92) In step **204**, when the new parent node is a root node, it generates and returns a first pairing approval packet to the network node, wherein a sending address of the first pairing approval packet is the MAC address of the new parent node, and a receiving address of the first pairing approval packet is the MAC address of the network node. The format of the first pairing approval packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames. In an embodiment, the step **204** of generating and returning, by the new parent node, the first pairing approval packet to the network node when the new parent node is the root node further comprises: filling, by the new parent node, an address 1 field in a null data frame with the MAC address of the network node, filling an address 2 field in the null data frame with the MAC address of the new parent node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a fourth preset value to define a type of the handshake frame as pairing approval, so as to generate and return the first pairing approval packet.
(93) In step **205**, when the new parent node is not the root node, it forwards the pairing change request packet upstream after changing the receiving address the pairing change request packet to a MAC address of its parent node.
(94) In step **208**, if the timer does not expire and the network node receives the first pairing approval packet, it means that the pairing switching is successful.
(95) In an embodiment, when the new parent node is not the root node, after the step **205**, the method for switching pairing in the mesh network system may further comprise: step **206**, step **207** and step **208***a*.
(96) In step **206**, after the relay node receiving the pairing change request packet determines that it can provide relay services for the network node and its descendant node, it adds the MAC addresses of the network node and its descendant node to a refugee table stored by itself, changes the receiving address of the pairing change request packet to a MAC address of its parent node, and then forwards the pairing change request packet upstream until a root node corresponding to the new parent node receives the pairing change request packet, so that the root node corresponding to the new parent node adds the MAC addresses of the network node and its descendant node to a refugee table stored by itself after determining that it can provide relay services for the network node and its descendant node, and then generates and sends a second pairing approval packet downstream. A sending address of the second pairing approval packet is a MAC address of the root

node corresponding to the new parent node, and a receiving address of the second pairing approval packet is the MAC address of the network node. The format of the second pairing approval packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames. In an embodiment, the step **206** of generating and sending, by the root node corresponding to the new parent node, the second pairing approval packet downstream comprises: filling, by the root node corresponding to the new parent node, an address 1 field in a null data frame with the MAC address of the network node, filling an address 2 field in the null data frame with the MAC address of the root node corresponding to the new parent node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a fourth preset value to define a type of the handshake frame as pairing approval, so as to generate and send the second pairing approval packet.
(97) In step **207**, when a relay node receiving the second pairing approval packet determines that the sending address of the second pairing approval packet is a MAC address of its parent node and the receiving address of the second pairing approval packet is a MAC address contained in a refugee table stored by itself, it changes the sending address of the second pairing approval packet to a MAC address of itself, and then forwards the second pairing approval packet downstream until the network node receives the second pairing approval packet.
(98) In step **208***a*, if the timer does not expire and the network node receives the second pairing approval packet, it means that the pairing switching is successful.
(99) In one embodiment, after the above-mentioned step **201**, the method for switching pairing in the mesh network system may further comprise step **202***a*. In step **202***a*, the network node is set to only receive various types of packets that meet a filtering condition, wherein the filtering condition is that a receiving address of an uplink packet is the MAC address of the network node, or a sending address of a packet is the MAC address of the new parent node and a receiving address of the packet is the MAC address of the network node.
(100) In one embodiment, in addition to the above step **201** to step **208** and step **208***a*, the method for switching pairing in the mesh network system may further comprise: if the timer expires, it means that the pairing switching fails. Specifically, if the timer expires and the network node does not receive the first pairing approval packet or the second pairing approval packet, it means that the pairing switching fails.
(101) In one embodiment, in addition to the above step **201** to step **208** and step **208***a*, the method for switching pairing in the mesh network system may further comprise: generating and returning, by the new parent node receiving the pairing change request packet, a pairing rejection packet to the network node when determining that it cannot provide relay services for the network node and its descendant node, wherein a sending address of the pairing rejection packet is a MAC address of itself (that is, the MAC address of the new parent node), and a receiving address of the pairing rejection packet is the MAC address of the network node; and the pairing switching failing when the network node receives the pairing rejection packet.
(102) In one embodiment, in addition to the above step **201** to step **208** and step **208***a*, the method for switching pairing in the mesh network system may further comprise: generating and sending, by any one relay node receiving the pairing change request packet or the root node corresponding to the new parent node, a pairing rejection packet when determining that it cannot provide relay services for the network node and its descendant node; and the pairing switching failing when the network node receives the pairing rejection packet. A sending address of the pairing rejection packet is a MAC address of itself, a receiving address of the pairing rejection packet is the MAC address of the network node, and the pairing rejection packet carries the MAC addresses of the network node and its descendant node. The relay node receiving the pairing rejection packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the pairing rejection packet to a MAC address of itself, and

then forwarding the pairing rejection packet downstream until the network node receives the pairing rejection packet.

(103) Please refer to FIG. **19** and FIG. **20**, which are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **17** and FIG. **18**. The mesh network system **3100** comprises an access point **3110** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprises a plurality of relay nodes (that is, a relay nodes **3122***a*, a relay node **3122***b* and a relay node **3130**), the relay node **3122***b* and the relay node **3130** serve as the root nodes communicating with the access point **3110**, and the parent node of the relay node **3122***a* is the relay node **3122***b*. In the process of constructing the mesh network system **3100** based on the above method for pairing nodes, each of the relay node **3122***a*, the relay node **3122***b* and the relay node **3130** stores the MAC address(es) of its descendant node(s) in the refugee table stored by itself.

(104) In FIG. **19**, the parent node of the relay node **3130** is the access point **3110**. In FIG. **20**, the parent node of the relay node **3130** is changed to the relay node **3122***a*. The whole process of changing the parent node of the relay node **3130** from the access point **3110** to the relay node **3122***a* is as follows.

(105) When the relay node **3130** detects that the communication quality with the access point **3110** is poor, it performs a scanning procedure to select the relay node **3122***a*, so that the access point **3110** becomes the old parent node.

(106) The relay node **3130** is set to only receive an uplink packet whose receiving address is a MAC address of itself (that is, the MAC address of the relay node **3130**), or a packet with a sending address being the MAC address of the relay node **3122***a* and a receiving address being a MAC address of itself (that is, the MAC address of the relay node **3130**).

(107) Since the relay node **3122***a* (i.e., the new parent node) is not an access point, the relay node **3130** generates and sends a pairing change request packet carrying the MAC address of its descendant node, the MAC address of the relay node **3130**, and the MAC address of the access point **3110** to the relay node **3122***a*, and starts a timer, wherein the sending address of the pairing change request packet is the MAC address of the relay node **3130**, and the receiving address of the pairing change request packet is the MAC address of the relay node **3122***a*. It should be noted that, since the relay node **3130** is changed from a root node to a non-root node during pairing switching, the relay node **3130** becomes the old root node when pairing is switched.

(108) After the relay node **3122***a* receives the pairing change request packet, it first determines whether it can provide the relay services for the relay node **3130** and its descendant node. When the relay node **3122***a* determines that it can provide the relay services for the relay node **3130** and its descendant node, it adds the MAC addresses of the relay node **3130** and its descendant node to the refugee table stored by itself, and then determines that it is not a root node. Therefore, the relay node **3122***a* changes the receiving address of the received pairing change request packet from the MAC address of the relay node **3122***a* to the MAC address of its parent node, and then forwards the pairing change request packet upstream.

(109) After the relay node **3122***b* receives the pairing change request packet, it first determines whether it can provide the relay services for the relay node **3130** and its descendant node. When the relay node **3122***b* determines that it can provide the relay services for the relay node **3130** and its descendant node, it adds the MAC addresses of the relay node **3130** and its descendant node to the refugee table stored by itself, and then generates and sends a second pairing approval packet downstream since the relay node **3122***b* is a root node, wherein a sending address of the second pairing approval packet is the MAC address of the relay node **3122***b*, and a receiving address of the second pairing approval packet is the MAC address of the relay node **3130**.

(110) After receiving the second pairing approval packet, the relay node **3122***a* first determines that the sending address of the second pairing approval packet is the MAC address of its parent node

and the receiving address of the second pairing approval packet is the MAC address contained in the refugee table stored by itself, and then changes the sending address of the received second pairing approval packet from the MAC address of the relay node **3122***b* to the MAC address of the relay node **3122***a*, and finally forwards the changed second pairing approval packet to the relay node **3130** downstream.
(111) When the timer does not expire and the relay node **3130** receives the second pairing approval packet, it means that the relay node **3130** has successfully paired with the relay node **3122***a*.
(112) Please refer to FIG. **21**, which is a flow chart of a method for switching pairing in a mesh network system according to a second embodiment of the present disclosure. In addition to step **201** to step **208** and step **208***a* in FIG. **17** and FIG. **18**, the method for switching pairing in the mesh network system may further comprise: step **210** to step **213**. It should be noted that, in order to avoid the drawing in FIG. **21** being too complicated, step **201** to step **208** and step **208***a* are omitted to be drawn.
(113) In step **210**, when the root node corresponding to the new parent node is not the same as the root node corresponding to the old parent node, the root node corresponding to the new parent node further generates and sends a pairing switching packet carrying a pairing switch identifier, the MAC addresses of the network node and its descendant node, and a MAC address of the old parent node to the access point, so that the access point forwards the pairing switching packet to the old root node. A receiving address of the pairing switching packet is the MAC address of the access point, a sending address of the pairing switching packet is a MAC address of itself (that is, the MAC address of the root node corresponding to the new parent node), and a destination address of the pairing switching packet is the MAC address of the old root node. The pairing switching packet can be but not limited to a normal packet defined by the 802.11 protocol or a QoS packet. The pairing switching packet may comprise: an address 1 field, an address 2 field, an address 3 field, and a payload field.
(114) In one embodiment, the step **210** of generating and sending, by the root node corresponding to the new parent node, the pairing switching packet carrying the pairing switch identifier, the MAC addresses of the network node and its descendant node, and the MAC address of the old parent node to the access point further comprises: filling, by the root node corresponding to the new parent node, an address 1 field in a data frame with the MAC address of the access point, filling an address 2 field in the data frame with the MAC address of the root node corresponding to the new parent node (that is, the MAC address of itself), filling an address 3 field in the data frame with the MAC address of the old root node, and filling a payload field in the data frame with the pairing switch identifier, the MAC addresses of the network node and its descendant node, and the MAC address of the old parent node, so as to generate and send the pairing switching packet to the access point.
(115) In one embodiment, after the network node confirms that the pairing switching is successful (that is, step **208** or step **208***a*), it additionally notifies the new root node to perform step **210**. In another embodiment, no matter whether the pairing switching is successful or not, the root node corresponding to the new parent node may generate and send the pairing switching packet to the access point while or after generating and sending the second pairing approval packet downstream.
(116) In step **211**, after the old root node receives the pairing switching packet, it deletes the MAC addresses of the network node and its descendant node from a refugee table stored by itself, and generates and sends a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, a sending address of the second pairing cancellation packet is the MAC address of the old root node, and the format of the second pairing cancellation packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames.
(117) In one embodiment, the format of the second pairing cancellation packet can expand a

payload field on the basis of a null data frame, so that the step **211** of generating and sending, by the old root node, the second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream comprise: filling, by the old root node, an address 1 field in a null data frame with the MAC address of the old parent node, filling an address 2 field in the null data frame with the MAC address of the old root node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, setting a plurality of second preset bits of the address 3 field in the null data frame to a fifth preset value to define a type of the handshake frame as a pairing cancellation, and filling a payload field in the null data frame, which is expanded, with the MAC addresses of the network node and its descendant node, so as to generate and send the second pairing cancellation packet downstream. In an example, the first preset bit of the address 3 field may be bit**0**, the first preset value may be 1 (that is, bit**0** is 1), the plurality of second preset bits of the address 3 field may be bit**1** to bit**4**, and the fifth preset value may be 3 (that is, bit**3** and bit**4** are 1, and bit**1** and bit**2** are 0).

(118) In step **212**, the relay node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet.

(119) In step **213**, after receiving the second pairing cancellation packet, the old parent node deletes the MAC addresses of the network node and its descendant node from a refugee table stored by itself.

(120) Therefore, the old parent node and its upper-level node(s) can delete the MAC addresses of the network node and its descendant node from the refugee table stored by itself in the process of pairing switching. That is, the old parent node and its upper-level node(s) do not provide relay services for the network node and its descendant node(s).

(121) Please refer to FIG. **22** and FIG. **23**, which are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **21**. The mesh network system **3200** comprises an access point **3210** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprise a plurality of relay nodes (i.e., a relay nodes **3212***a*, a relay node **3212***b*, a relay node **3222***a*, a relay node **3222***b* and a relay node **3230**). The relay node **3222***b* and the relay node **3212***b* serve as the root nodes communicating with the access point **3210**, and the parent node of the relay node **3212***a* is the relay node **3212***b*, the parent node of the relay node **3222***a* is the relay node **3222***b*. In the process of constructing the mesh network system **3200** based on the above-mentioned method for pairing nodes, each of the relay node **3212***a*, the relay node **3212***b*, the relay node **3222***a*, the relay node **3222***b*, and the relay node **3230** stores the MAC address(es) of its descendant node(s) in its refugee table.

(122) In FIG. **22**, the parent node of the relay node **3230** is the relay node **3212***a*. In FIG. **23**, the parent node of the relay node **3130** is changed to the relay node **3222***a*. The whole process of changing the parent node of the relay node **3230** from the relay node **3212***a* to the relay node **3222***a* is as follows.

(123) When the relay node **3230** detects that the communication quality with the relay node **3212***a* is poor or receives a first pairing cancellation packet from the relay node **3212***a*, it performs the scanning procedure to select the relay node **3222***a*, so that the relay node **3212***a* becomes the old parent node.

(124) The relay node **3230** is set to only receive an uplink packet whose receiving address is the MAC address of itself (that is, the MAC address of the relay node **3230**), or a packet with a sending address being the MAC address of the relay node **3222***a* and a receiving address being the MAC address of itself (that is, the MAC address of the relay node **3230**).

(125) Since the relay node **3222***a* is not an access point, the relay node **3230** generates and sends a pairing change request packet carrying a MAC address of its descendant node, the MAC address of the relay node **3230**, the MAC address of the relay node **3212***a*, and the MAC address of the relay node **3222***a* to the relay node **3222***a*, and starts a timer, wherein a sending address of the pairing change request packet is the MAC address of the relay node **3230** and a receiving address of the pairing change request packet is the MAC address of the relay node **3222***a*.
(126) After the relay node **3222***a* receives the pairing change request packet, it first determines whether it can provide relay services for the relay node **3230** and its descendant node. When the relay node **3222***a* determines that it can provide relay services for the relay node **3230** and its descendant node, it adds the MAC addresses of the relay node **3230** and its descendant node to the refugee table stored by itself, and then determines that it is not a root node. Therefore, the relay node **3222***a* changes the receiving address of the received pairing change request packet from the MAC address of the relay node **3222***a* to the MAC address of its parent node, and then forwards the changed pairing change request packet upstream.
(127) After the relay node **3222***b* receives the pairing change request packet, it first determines whether it can provide relay services for the relay node **3230** and its descendant node. When the relay node **3222***b* determines that it can provide relay services for the relay node **3230** and its descendant node, it adds the MAC addresses of the relay node **3230** and its descendant node to the refugee table stored by itself, and then generates and sends the second pairing approval packet downstream since the relay node **3222***b* is a root node, wherein a sending address of the second pairing approval packet is the MAC address of the relay node **3222***b*, and a receiving address of the second pairing approval packet is the MAC address of the relay node **3230**.
(128) After receiving the second pairing approval packet, the relay node **3222***a* first determines that the sending address of the second pairing approval packet is the MAC address of its parent node and the receiving address of the second pairing approval packet is the MAC address contained in the refugee table stored by itself, and then changes the sending address of the received second pairing approval packet from the MAC address of the relay node **3222***b* to the MAC address of the relay node **3222***a*, and finally forwards the changed second pairing approval packet to the relay node **3230** downstream.
(129) When the timer does not expire and the relay node **3230** receives the second pairing approval packet, it means that the relay node **3230** has successfully paired with the relay node **3222***a*.
(130) Since the root node corresponding to the relay node **3222***a* is different from the root node corresponding to the relay node **3212***a*, the relay node **3222***b* may generate and send a pairing switching packet carrying a pairing switch identifier, the MAC addresses of the relay node **3230** and its descendant node, and the MAC address of the relay node **3212***a* to the access point **3210** while or after generating and sending the second pairing approval packet downstream, so that the access point **3210** forwards the pairing switching packet to the relay node **3212***b*, wherein a receiving address of the pairing switching packet is the MAC address of the access point **3210**, a sending address of the pairing switching packet is the MAC address of the relay node **3222***b*, and a destination address of the pairing switching packet is the MAC address of the relay node **3212***b*.
(131) After the relay node **3212***b* receives the pairing switching packet, it deletes the MAC addresses of the relay node **3230** and its descendant node from the refugee table stored by itself, and generates and sends a second pairing cancellation packet carrying the MAC addresses of the relay node **3230** and its descendant node, and the MAC address of the relay node **3222***a* downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the relay node **3212***a* and a sending address of the second pairing cancellation packet is the MAC address of the relay node **3212***b*.
(132) After receiving the second pairing cancellation packet, the relay node **3212***a* deletes the MAC addresses of the relay node **3230** and its descendant node from the refugee table stored by itself.
(133) Please refer to FIG. **24**, which is a flow chart of a method for switching pairing in a mesh

network system according to a third embodiment of the present disclosure. In addition to step **201** to step **208** and step **208***a* in FIG. **17** and FIG. **18**, the method for switching pairing in the mesh network system may further comprise: step **225** to step **228**. It should be noted that, in order to avoid the drawing in FIG. **24** being too complicated, steps **201** to **208** and **208***a* are omitted to be drawn.

(134) In step **225**, when the root node corresponding to the new parent node is not the same as the root node corresponding to the old parent node, the root node corresponding to the new parent node further generates and sends a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node to the old root node, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the access point. The manner in which the root node corresponding to the new parent node generates the second pairing cancellation packet in step **225** may be the same as the manner in which the old root node generates the second pairing cancellation packet in step **211**, and details will not be repeated here.

(135) In one embodiment, after the network node confirms that the pairing switching is successful (that is, step **208** or step **208***a*), it additionally notifies the new root node to perform step **225**. In another embodiment, no matter whether the pairing switching is successful or not, the root node corresponding to the new parent node may generate and send the second pairing cancellation packet while or after generating and sending the second pairing approval packet downstream.

(136) In step **226**, the old root node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream.

(137) In step **227**, the relay node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet.

(138) In step **228**, after receiving the second pairing cancellation packet, the old parent node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself.

(139) Therefore, by the configuration of step **225**, the new parent node directly generates and sends the second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node to the old root node, instead of the step **210** of generating and sending, by the new parent node, the pairing switching packet to the access point, so that the access point forwards the pairing switching packet to the old root node, and the step **211** of generating and sending, by the old root node, the second pairing cancellation packet downstream.

(140) Please refer to FIG. **25**, which is a flow chart of a method for switching pairing in a mesh network system according to a fourth embodiment of the present disclosure. In addition to step **201** to step **208** and step **208***a* in FIG. **17** and FIG. **18**, the method for switching pairing in the mesh network system may further comprise: step **214** to step **216**. It should be noted that, in order to avoid the drawing in FIG. **24** being too complicated, step **201** to step **208** and step **208***a* are omitted to be drawn.

(141) In step **214**, when the root node corresponding to the new parent node is the same as the root node corresponding to the old parent node, the root node corresponding to the new parent node generates and sends a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of itself; the format of the second pairing cancellation packet can be a frame defined by the 802.11 protocol, such as a data frame, a

management frame, a control frame, and other self-defined frames; and the manner in which the root node corresponding to the new parent node generates the second pairing cancellation packet in step **214** may be the same as the manner in which the old root node generates the second pairing cancellation packet in step **211**, and details will not be repeated here.

(142) In one embodiment, after the network node confirms that the pairing switching is successful (that is, step **208** or step **208***a*), it additionally notifies the new root node to perform step **214**. In another embodiment, no matter whether the pairing switching is successful or not, the root node corresponding to the new parent node may generate and send the second pairing cancellation packet downstream while or after generating and sending the second pairing approval packet downstream.

(143) In step **215**, the relay node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet.

(144) In step **216**, after the old parent node receives the second pairing cancellation packet, it deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself.

(145) Therefore, the old parent node and its upper-level node(s) can delete the MAC addresses of the network node and its descendant node from the refugee table stored by itself in the process of pairing switching. That is, the old parent node and its upper-level node(s) do not provide relay services for the network node and its descendant node(s). In addition, since the root node corresponding to the new parent node is the same as the root node corresponding to the old parent node in this embodiment, the root node corresponding to the new parent node does not need to generate a pairing switching packet to the access point in this embodiment.

(146) Please refer to FIG. **26** and FIG. **27**, which are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **25**. The mesh network system **3300** comprises an access point **3310** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprise a plurality of relay nodes (that is, a relay nodes **3312***a*, a relay node **3312***b*, a relay node **3312***c*, a relay node **3322***a*, and a relay node **3330**). The relay node **3312***c* serves as the root node communicating with the access point **3310**, the parent node of the relay node **3312***a* is the relay node **3312***b*, the parent node of the relay node **3312***b* is the relay node **3312***c*, and the parent node of the relay node **3322***a* is the relay node **3312***c*. In the process of constructing the mesh network system **3300** based on the above method for pairing nodes, each of the relay node **3312***a*, the relay node **3312***b*, the relay node **3312***c*, the relay node **3322***a*, and the relay node **3330** stores the MAC address(es) of its descendant node(s) in its refugee table.

(147) In FIG. **26**, the parent node of the relay node **3330** is the relay node **3312***a*. In FIG. **27**, the parent node of the relay node **3330** is changed to the relay node **3322***a*. The whole process of changing the parent node of the relay node **3330** from the relay node **3312***a* to the relay node **3322***a* is as follows.

(148) When the relay node **3330** detects that the communication quality with the relay node **3312***a* is poor or receives the first pairing cancellation packet from the relay node **3312***a*, it performs a scanning procedure to select the relay node **3322***a*, so that the relay node **3312***a* becomes the old parent node.

(149) The relay node **3330** is configured to only receive an uplink packet whose receiving address is the MAC address of the relay node **3330**, or a packet with a sending address being the MAC address of the relay node **3322***a* and a receiving address being the MAC address of the relay node **3330**.

(150) Since the relay node **3322***a* is not an access point, the relay node **3330** generates and sends a

pairing change request packet carrying the MAC address of its descendant node, the MAC address of the relay node **3330**, and the MAC address of the relay node **3312***a* to the relay node **3322***a*, and starts a timer, wherein a sending address of the pairing change request packet is the MAC address of the relay node **3330** and a receiving address of the pairing change request packet is the MAC address of the relay node **3322***a*.
(151) After the relay node **3322***a* receives the pairing change request packet, it first determines whether it can provide the relay services for the relay node **3330** and its descendant node. When the relay node **3322***a* determines that it can provide the relay services for the relay node **3330** and its descendant node, it adds the MAC addresses of the relay node **3330** and its descendant node to the refugee table stored by itself, and then determines that it is not the root node. Therefore, the relay node **3322***a* changes the receiving address of the received pairing change request packet from the MAC address of the relay node **3322***a* to the MAC address of its parent node, and then forwards the changed pairing change request packet upstream.
(152) After the relay node **3312***c* receives the pairing change request packet, it first determines whether it can provide the relay services for the relay node **3330** and its descendant node. When the relay node **3312***c* determines that it can provide the relay services for the relay node **3330** and its descendant node, it adds the MAC addresses of the relay node **3330** and its descendant node to the refugee table stored by itself, and then generates and sends the second pairing approval packet downstream since the relay node **3312***c* is the root node, wherein the sending address of the second pairing approval packet is the MAC address of the relay node **3312***c*, and the receiving address of the second pairing approval packet is the MAC address of the relay node **3330**.
(153) After receiving the second pairing approval packet, the relay node **3322***a* first determines that the sending address of the second pairing approval packet is the MAC address of its parent node and the receiving address is the MAC address contained in the refugee table stored by itself, and then changes the sending address of the received second pairing approval packet from the MAC address of the relay node **3312***c* to the MAC address of the relay node **3322***a*, and finally forwards the changed second pairing approval packet to the relay node **3330** downstream.
(154) When the timer does not expire and the relay node **3330** receives the second pairing approval packet, it means that the relay node **3330** has successfully paired with the relay node **3322***a*.
(155) Since the root node corresponding to the relay node **3322***a* is the same as the root node corresponding to the relay node **3312***a*, the relay node **3312***c* generates and sends the second pairing cancellation packet, which carries the MAC addresses of the relay node **3330** and its descendant node and the MAC address of the relay node **3322***a*, downstream while or after it generates and sends the second pairing approval packet downstream. A receiving address of the second pairing cancellation packet is the MAC address of the relay node **3312***a*, and a sending address of the second pairing cancellation packet is the MAC address of the relay node **3312***c*.
(156) After the relay node **3312***b* receives the second pairing cancellation packet, it deletes the MAC addresses of the relay node **3330** and its descendant node from the refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of the relay node **3312***b*, and then forwards the second pairing cancellation packet downstream to the relay node **3312***a*.
(157) After receiving the second pairing cancellation packet, the relay node **3312***a* deletes the MAC addresses of the relay node **3330** and its descendant node from the refugee table stored by itself.
(158) Please refer to FIG. **28**, which is a flow chart of a method for switching pairing in a mesh network system according to a fifth embodiment of the present disclosure. In addition to step **201** to step **208** and step **208***a* in FIG. **17** and FIG. **18**, the method for switching pairing in the mesh network system may further comprise: step **217** to step **219**. It should be noted that, in order to avoid the drawing of FIG. **27** being too complicated, step **201** to step **208** and step **208***a* are omitted to drawn; and the pairing change request packet described in step **202** in this embodiment carries the MAC address of the descendant node of the network node, the MAC address of the old root

node, the MAC address of the old parent node, and the MAC address of the new parent node, so as to facilitate the performance of the subsequent steps.
(159) In step **217**, when the root node corresponding to the new parent node is the same as the root node corresponding to the old parent node, the root node corresponding to the new parent node generates and sends a third pairing cancellation packet carrying the MAC addresses of the network node and its descendant node and the MAC address of the new parent node downstream, wherein a receiving address of the third pairing cancellation packet is the MAC address of the old parent node, and a sending address is a MAC address of itself; the format of the third pairing cancellation packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames; and the manner in which the root node corresponding to the new parent node generates the third pairing cancellation packet in step **217** is similar to the manner in which the old root node generates the second pairing cancellation packet in above step **211**, and the difference between the two manners is that the expanded payload field of the third pairing cancellation packet is further filled with the MAC address of the new parent node.
(160) In one embodiment, after the network node confirms that the pairing switching is successful (that is, step **208** or step **208***a*), it additionally notifies the new root node to perform step **217**. In another embodiment, no matter whether the pairing switching is successful or not, the root node corresponding to the new parent node may generate and send the third pairing cancellation packet downstream while or after generating and sending the third pairing approval packet downstream.
(161) In step **218**, the relay node that receives the third pairing cancellation packet first determines whether the MAC address of itself is the same as the MAC address of the new parent node or determines whether the MAC address of the new parent node is contained in the refugee table stored by itself, and then forwards the third pairing cancellation packet downstream until the old parent node receives the third pairing cancellation packet, wherein if it determines that the MAC address of itself is the same as the MAC address of the new parent node or the MAC address of the new parent node is contained in the refugee table stored by itself, the relay node changes the sending address of the third pairing cancellation packet to the MAC address of itself, and then forwards the third pairing cancellation packet downstream; if it determines that the MAC address of itself is different from the MAC address of the new parent node or the MAC address of the new parent node is not contained in the refugee table stored by itself, the relay node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the third pairing cancellation packet to the MAC address of itself, and then forwards the third pairing cancellation packet downstream.
(162) In step **219**, after receiving the third pairing cancellation packet, the old parent node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself.
(163) Therefore, by the configuration of step **218**, the situation where the new parent node and its upper-level node(s) mistakenly delete the MAC addresses of the network node and its descendant node from the refugee table stored by itself when the new parent node is the upper-level node of the old parent node is avoided.
(164) Please refer to FIG. **29** and FIG. **30**, which are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **28**. The mesh network system **3400** comprises an access point **3410** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprises a plurality of relay nodes (that is, a relay nodes **3412***a*, a relay node **3412***b*, a relay node **3412***c*, and a relay node **3430**). The relay node **3412***c* serves as the root node communicating with the access point **3410**, the parent node of the relay node **3412***a* is the relay node **3412***b*, and the parent node of the relay node **3412***b* is the relay node **3412***c*. In the process of constructing the mesh network system **3400** based on the above-mentioned method for pairing nodes, each of the relay node **3412***a*, the relay node **3412***b*, the relay node **3412***c* and the

relay node **3430** stores the MAC address(es) of its descendant node(s) in its refugee table.
(165) In FIG. **29**, the parent node of the relay node **3430** is the relay node **3412***a*. In FIG. **30**, the parent node of the relay node **3430** is changed to the relay node **3412***b*. The whole process of changing the parent node of relay node **3430** from relay node **3412***a* to relay node **3412***b* is as follows.
(166) When the relay node **3430** detects that the communication quality with the relay node **3412***a* is poor or receives the first pairing cancellation packet from the relay node **3412***a*, it performs a scanning procedure to select the relay node **3412***b*, so that the relay node **3412***a* becomes the old parent node.
(167) The relay node **3430** is configured to only receive an uplink packet whose receiving address is the MAC address of the relay node **3430**, or a packet with a sending address being the MAC address of the relay node **3412***b*, and a receiving address being the MAC address of the relay node **3430**.
(168) Since the relay node **3412***b* is not an access point, the relay node **3430** generates and sends a pairing change request packet carrying the MAC address of its descendant node, the MAC address of the relay node **3430**, the MAC address of the relay node **3412***a*, and the MAC address of the relay node **3412***c* to the relay node **3412***b*, and starts a timer, wherein a sending address of the pairing change request packet is the MAC address of the relay node **3430**, and a receiving address of the pairing change request packet is the MAC address of the relay node **3412***b*.
(169) After the relay node **3412***b* receives the pairing change request packet, it first determines whether it can provide the relay services for the relay node **3430** and its descendant node. When the relay node **3412***b* determines that it can provide the relay services for the relay node **3430** and its descendant node, it adds the MAC addresses of the relay node **3430** and its descendant node to the refugee table stored by itself, and then determines that it is not a root node. Therefore, the relay node **3412***b* changes the receiving address of the received pairing change request packet from the MAC address of the relay node **3412***b* to the MAC address of its parent node, and then forwards the changed pairing change request packet upstream.
(170) After the relay node **3412***c* receives the pairing change request packet, it first determines whether it can provide relay services for the relay node **3430** and its descendant node. When the relay node **3312***c* determines that it can provide the relay services for the relay node **3430** and its descendant node, it adds the MAC addresses of the relay node **3430** and its descendant node to the refugee table stored by itself, and then generates and sends the second pairing approval packet downstream since the relay node **3412***c* is a root node, wherein a sending address of the second pairing approval packet is the MAC address of the relay node **3412***c*, and a receiving address of the second pairing approval packet is the MAC address of the relay node **3430**.
(171) After receiving the second pairing approval packet, the relay node **3412***b* first determines that the sending address of the second pairing approval packet is the MAC address of its parent node and the receiving address of the second pairing approval packet is the MAC address contained in the refugee table stored by itself, and then changes the sending address of the received second pairing approval packet from the MAC address of the relay node **3412***c* to the MAC address of the relay node **3412***b*, and finally forwards the changed second pairing approval packet to the relay node **3430** downstream.
(172) When the timer does not expire and the relay node **3430** receives the second pairing approval packet, it means that the relay node **3430** has successfully paired with the relay node **3412***b*.
(173) Since the root node corresponding to the relay node **3412***b* is the same as the root node corresponding to the relay node **3412***a*, the relay node **3412***c* generates and sends a third pairing cancellation packet, which carries the MAC addresses of the relay node **3430** and its descendant node and the MAC address of the relay node **3412***b*, downstream while or after it generates and sends the second pairing approval packet downstream, wherein a receiving address of the third pairing cancellation packet is the MAC address of the relay node **3412***a* and a sending address of

the third pairing cancellation packet is the MAC address of the relay node **3412***c*.
(174) After the relay node **3412***b* receives the third pairing cancellation packet, it determines that it is the new parent node (that is, it determines that the MAC address of the new parent node (relay node **3412***b*) of the third pairing cancellation packet is the MAC address of itself), and changes the sending address to the MAC address of the relay node **3412***b*, and then forwards the third pairing cancellation packet downstream.
(175) After receiving the third pairing cancellation packet, the relay node **3412***a* deletes the MAC addresses of the relay node **3430** and its descendant node from the refugee table stored by itself.
(176) Please refer to FIG. **31**, which is a flow chart of a method for switching pairing in a mesh network system according to a sixth embodiment of the present disclosure. As shown in FIG. **31**, the method for switching pairing in the mesh network system comprises: step **220** to step **224**.
(177) In step **220**, a network node performs a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node. Step **220** is the same as step **201**, and details will not be repeated here.
(178) In step **221**, when the new parent node is an access point, the network node generates and sends a pairing switch packet carrying a pairing switch identifier, MAC addresses of the network node and its descendant node, and a MAC address of the old parent node to the access point, so that the access point forwards the pairing switching packet to an old root node, wherein a receiving address of the pairing switching packet is a MAC address of the access point, a sending address of the pairing switching packet is a MAC address of the network node, and a destination address of the pairing switching packet is a MAC address of the old root node. The pairing switching packet can be, but not limited to, a normal packet defined by 802.11 protocol or a QoS packet. The manner in which the network node generates the pairing switching packet in step **221** is the same as the manner in which the root node corresponding to the new parent node generates the pairing switching packet in step **210**, and details will not be repeated here.
(179) In step **222**, after the old root node receives the pairing switching packet, it deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, and generates and sends a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the old root node. The format of the second pairing cancellation packet can be a frame defined by the 802.11 protocol, such as a data frame, a management frame, a control frame, and other self-defined frames. The manner in which the old root node generates the second pairing cancellation packet in step **222** is the same as the manner in which the old root node generates the second pairing cancellation packet in step **211**, and details will not be repeated here.
(180) In step **223**, the relay node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, and changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet.
(181) In step **224**, after receiving the second pairing cancellation packet, the old parent node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself.
(182) In one embodiment, after the above-mentioned step **220**, the method for switching pairing in the mesh network system may further comprise: setting the network node to only receive various types of packets that meet a filtering condition, wherein the filtering condition is that a receiving address of an uplink packet is the MAC address of the network node, or a sending address of a packet is the MAC address of the new parent node and a receiving address of the packet is the MAC address of the network node (step **221***a*).

(183) Please refer to FIG. **32** and FIG. **33**, which are schematic diagrams of an embodiment of the mesh network system before and after applying the method for switching pairing in FIG. **31**. The mesh network system **3500** comprises an access point **3510** and a plurality of network nodes, the plurality of network nodes can communicate with each other through the mesh network, and the plurality of network nodes comprise a plurality of relay nodes (that is, a relay node **3512***a*, a relay node **3512***b* and a relay node **3530**). The relay node **3512***b* serves as the root node communicating with the access point **3510**, and the parent node of the relay node **3512***a* is the relay node **3512***b*. In the process of constructing the mesh network system **3500** based on the above-mentioned method for pairing nodes, each of the relay node **3512***a*, the relay node **3512***b* and the relay node **3530** stores the MAC address(es) of its descendant node(s) in its refugee table.
(184) In FIG. **32**, the parent node of the relay node **3530** is the relay node **3512***a*. In FIG. **33**, the parent node of the relay node **3530** is changed to the access point **3510**, and the relay node **3530** is changed to the new root node. The whole process of changing the parent node of the relay node **3530** from the relay node **3512***a* to the access point **3510** is as follows.
(185) When the relay node **3530** detects that the communication quality with the relay node **3512***a* is poor or receives the first pairing cancellation packet from the relay node **3512***a*, it performs the scanning procedure to select the access point **3510**, so that the relay node **3512***a* becomes the old parent node.
(186) The relay node **3530** is configured to only receive an uplink packet whose receiving address is the MAC address of the relay node **3530**, or a packet with a sending address being the MAC address of the access point **3510** and a receiving address being the MAC address of the relay node **3530**.
(187) Since the new parent node of the relay node **3530** is an access point, the relay node **3530** generates and sends a pairing switching packet carrying a pairing switch identifier, the MAC addresses of the relay node **3530** and its descendant node, and the MAC address of the relay node **3512***a* to the access point **3510**, so that the access point **3510** forwards the pairing switching packet to the relay node **3512***a*, wherein a receiving address of the pairing switching packet is the MAC address of the access point **3510**, a sending address of the pairing switching packet is the relay node **3530**, and a destination address of the pairing switching packet is the MAC address of relay node **3512***b*.
(188) After the relay node **3512***b* receives the pairing switching packet, it deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, and generates and sends a second pairing cancellation packet carrying the MAC addresses of the relay node **3530** and its descendant node and the MAC address of the access point **3510** downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the relay node **3512***a*, and a sending address of the second pairing cancellation packet is the MAC address of the relay node **3512***b*.
(189) The relay node **3512***a* that receives the second pairing cancellation packet deletes the MAC addresses of the relay node **3530** and its descendant node from the refugee table stored by itself.
(190) Please refer to FIG. **34**, which is a flow chart of a method for switching pairing in a mesh network system according to a seventh embodiment of the present disclosure. As shown in FIG. **34**, the method for switching pairing in the mesh network system comprises: step **229** to step **234**.
(191) In step **229**, a network node performs a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node. Step **229** is the same as step **201**, and details will not be repeated here.
(192) In step **230**, when the new parent node is not an access point, the network node generates and sends a pairing change request packet carrying a MAC address of its descendant node, a MAC address of an old root node and a MAC address of the old parent node to the new parent node, and starts a timer, wherein a sending address of the pairing change request packet is a MAC address of the network node, and a receiving address of the pairing change request packet is a MAC address

of the new parent node, and the old root node is a root node corresponding to the old parent node. Step **220** is the same as step **202**, and details will not be repeated here.
(193) In step **231**, the new parent node that receives the pairing change request packet determines that the MAC addresses of the network node and its descendant node are contained in the refugee table stored by itself, and then generates and returns a first pairing approval packet to the network node, and generates and sends a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant nodes downstream, wherein the refugee table stored by the new parent node comprises the MAC addresses of all subordinate nodes that the new parent node can provide relay services for, a sending address of the first pairing approval packet is the MAC address of the new parent node, a receiving address of the first pairing approval packet is the MAC address of the network node, a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the new parent node. The manner in which the root node corresponding to the new parent node generates the second pairing cancellation packet in step **231** is the same as the manner in which the old root node generates the second pairing cancellation packet in step **211**, and details will not be repeated here.
(194) In this embodiment, since the new parent node is an upper-level node of the old parent node, the new parent node can generate the first pairing approval packet and the second pairing cancellation packet, without the situation where the root node corresponding to the new parent node generates the first pairing approval packet and the second pairing cancellation packet.
(195) In step **232**, if the network node receives the first pairing approval packet, it means that the pairing switching is successful.
(196) In step **233**, the relay node that receives the second pairing cancellation packet deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the second pairing cancellation packet to the MAC address of itself, and then forwards the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet.
(197) In step **234**, after receiving the second pairing cancellation packet, the old parent node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself.
(198) In one embodiment, after the above-mentioned step **229**, the method for switching pairing in the mesh network system may further comprise: setting the network node to only receive various types of packets that meet a filter condition, wherein the filtering condition is that a receiving address of an uplink packet is the MAC address of the network node, or a sending address of a packet is the MAC address of the new parent node and a receiving address of the packet is the MAC address of the network node (step **230***a*).
(199) It should be noted that, if there is no causal relationship between the above steps, the present disclosure does not limit the order of execution thereof.
(200) In summary, in the embodiments of the present disclosure, the method for switching pairing in the mesh network system can be applied to a mesh network that carries out packet forwarding based on the MAC address. By the design of the pairing switching packet, the pairing change request packet, the first pairing approval packet, the second pairing approval packet, the first pairing cancellation packet and the second pairing cancellation packet, the corresponding processing logic is proposed for the nodes in the mesh network system to complete the update of the refugee tables of the relevant nodes in the mesh network during the pairing switching process, which can realize fast, stable and efficient pairing switch.
(201) Although the present disclosure is disclosed in the foregoing embodiments, it is not intended to limit the present disclosure. Changes and modifications made without departing from the spirit and scope of the present disclosure belong to the scope of the claims of the present disclosure. The scope of protection of the present disclosure should be construed based on the following claims.

Claims

1. A method for switching pairing in a mesh network system, comprising: performing, by a network node, a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node; generating and sending, by the network node, a pairing change request packet carrying a MAC address of a descendant node of the network node, a MAC address of an old root node, and a MAC address of the old parent node to the new parent node when the new parent node is not an access point, wherein a sending address of the pairing change request packet is a MAC address of the network node, a receiving address of the pairing change request packet is a MAC address of the new parent node, and the old root node is a root node corresponding to the old parent node; generating and returning, by the new parent node receiving the pairing change request packet, a first pairing approval packet to the network node after determining that the MAC addresses of the network node and its descendant node are contained in a refugee table stored by itself, and generating and sending a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein the refugee table stored by the new parent node contains MAC addresses of all subordinate nodes that the new parent node provides relay services for, a sending address of the first pairing approval packet is the MAC address of the new parent node, a receiving address of the first pairing approval packet is the MAC address of the network node, a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the new parent node; wherein pairing switching is successful if the network node receives the first pairing approval packet; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant nodes from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.
2. The method according to claim 1, further comprising: generating and returning, by the new parent node, a first pairing approval packet to the network node when the new parent node is a root node, wherein a sending address of the first pairing approval packet is the MAC address of the new parent node and a receiving address of the first pairing approval packet is the MAC address of the network node; and forwarding, by the new parent node, the pairing change request packet upstream after changing the receiving address the pairing change request packet to a MAC address of its parent node when the new parent node is not the root node; wherein pairing switching is successful when the network node receives the first pairing approval packet.
3. The method according to claim 1, wherein the step of generating and sending, by the network node, the pairing change request packet carrying the MAC address of its descendant node, the MAC address of the old root node, and the MAC address of the old parent node to the new parent node further comprises: filling, by the network node, an address 1 field in a null data frame with the MAC address of the new parent node, filling an address 2 field in the null data frame with the MAC address of the network node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, setting a plurality of second preset bits of the address 3 field in the null data frame to a third preset value to define a type of the handshake frame as a pairing request, and filling an payload field in the null data frame, which is expanded, with the MAC address of the descendant node of the network node, the MAC address of the old root node, and the MAC address of the old parent node, so as to generate and send the pairing change request packet to the new parent node.
4. The method according to claim 2, wherein the step of generating and returning, by the new

parent node, the first pairing approval packet to the network node when the new parent node is the root node further comprises: filling, by the new parent node, an address 1 field in a null data frame with the MAC address of the network node, filling an address 2 field in the null data frame with the MAC address of the new parent node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a fourth preset value to define a type of the handshake frame as pairing approval, so as to generate and return the first pairing approval packet.

5. The method according to claim 1, further comprising: generating and returning, by the new parent node receiving the pairing change request packet, a pairing rejection packet to the network node when determining that it cannot provide relay services for the network node and its descendant node, wherein a sending address of the pairing rejection packet is a MAC address of itself, and a receiving address of the pairing rejection packet is the MAC address of the network node; wherein pairing switching is failure when the network node receives the pairing rejection packet.

6. The method according to claim 1, wherein after the step of performing, by the network node, the scanning procedure to select the new parent node, so that the current parent node becomes the old parent node, the method for switching pairing in the mesh network system further comprises: setting the network node to only receive various types of packets that meet a filtering condition, wherein the filtering condition is that a receiving address of an uplink packet is the MAC address of the network node, or a sending address of a packet is the MAC address of the new parent node and a receiving address of the packet is the MAC address of the network node.

7. The method according to claim 1, wherein the step of performing, by the network node, the scanning procedure to select the new parent node, so that the current parent node becomes the old parent node further comprises: obtaining, by the network node, communication information of each relay node in the mesh network system based on a plurality of beacon packets or a plurality of probe response packets obtained through the scanning procedure, wherein the communication information of each relay node comprises a signal strength between each relay node and the access point, a signal strength between the network node and each relay node, a level of each relay node in the mesh network system, relay service capability of each relay node, and/or a MAC address of each relay node; and selecting, by the network node, a relay node as the new parent node based on the communication information of each relay node.

8. The method according to claim 1, wherein when the network node counts that an uplink packet transmission failure rate reaches a first threshold, energy of a packet received from the current parent node is lower than a second threshold, or a beacon is continuously lost for a period of time, it represents that the network node detects that communication quality with the current parent node is poor, so that the network node performs the scanning procedure to select the new parent node; and when the current parent node detects that communication quality with the network node is degraded or receives a connection replacement instruction, the current parent node outputs a first pairing cancellation packet to the network node, so that the network node performs the scanning procedure to select the new parent node.

9. The method according to claim 1, further comprising: adding, by a relay node receiving the pairing change request packet, the MAC addresses of the network node and its descendant node to a refugee table stored by itself after determining that it can provide relay services for the network node and its descendant node, changing the receiving address of the pairing change request packet to a MAC address of its parent node, and then forwarding the pairing change request packet upstream until a root node corresponding to the new parent node receives the pairing change request packet, so that the root node corresponding to the new parent node adds the MAC addresses of the network node and its descendant node to a refugee table stored by itself after determining that it can provide relay services for the network node and its descendant node, and then generates

and sends a second pairing approval packet downstream, wherein a sending address of the second pairing approval packet is a MAC address of the root node corresponding to the new parent node, and a receiving address of the second pairing approval packet is the MAC address of the network node; and changing, by the relay node receiving the second pairing approval packet, the sending address of the second pairing approval packet to a MAC address of itself when determining that the sending address of the second pairing approval packet is the MAC address of its parent node and the receiving address of the second pairing approval packet is a MAC address contained in a refugee table stored by itself, and then forwarding the second pairing approval packet downstream until the network node receives the second pairing approval packet; wherein pairing switching is successful if the network node receives the second pairing approval packet.

10. The method according to claim 9, further comprising: generating and sending, by the root node corresponding to the new parent node, a pairing switching packet carrying a pairing switch identifier, the MAC addresses of the network node and its descendant node, and a MAC address of the old parent node to the access point when the root node corresponding to the new parent node is different from a root node corresponding to the old parent node, so that the access point forwards the pairing switching packet to the old root node, wherein a receiving address of the pairing switching packet is a MAC address of the access point, a sending address of the pairing switching packet is the MAC address of the root node corresponding to the new parent node, and a destination address of the pairing switching packet is the MAC address of the old root node; deleting, by the old root node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the pairing switching packet, and generating and sending a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the old root node; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.

11. The method according to claim 10, wherein the step of generating and sending, by the old root node, the second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream further comprises: filling, by the old root node, an address 1 field in a null data frame with the MAC address of the old parent node, filling an address 2 field in the null data frame with the MAC address of the old root node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, setting a plurality of second preset bits of the address 3 field in the null data frame to a fifth preset value to define a type of the handshake frame as a pairing cancellation, and filling a payload field in the null data frame, which is expanded, with the MAC addresses of the network node and its descendant node, so as to generate and send the second pairing cancellation packet downstream.

12. The method according to claim 10, wherein the step of generating and sending, by the root node corresponding to the new parent node, the pairing switching packet carrying the pairing switch identifier, the MAC addresses of the network node and its descendant node, and the MAC address of the old parent node to the access point further comprises: filling, by the root node corresponding to the new parent node, an address 1 field in a data frame with the MAC address of the access point, filling an address 2 field in the data frame with the MAC address of the root node corresponding to the new parent node, filling an address 3 field in the data frame with the MAC address of the old root node, and filling a payload field in the data frame with the pairing switch

identifier, the MAC addresses of the network node and its descendant node, and the MAC address of the old parent node, so as to generate and send the pairing switching packet to the access point.
13. The method according to claim 9, further comprising: generating and sending, by the root node corresponding to the new parent node, a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream when the root node corresponding to the new parent node is the same as a root node corresponding to the old parent node, wherein a receiving address of the second pairing cancellation packet is a MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the root node corresponding to the new parent node; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.
14. The method according to claim 9, further comprising: generating and sending, by the root node corresponding to the new parent node, a third pairing cancellation packet carrying the MAC addresses of the network node and its descendant node and the MAC address of the new parent node when the root node corresponding to the new parent node is the same as a root node corresponding to the old parent node, wherein a receiving address of the third pairing cancellation packet is the MAC address of the old parent node, and a sending address of the third pairing cancellation packet is the root node corresponding to the new parent node; determining, by a relay node receiving the third pairing cancellation packet, whether a MAC address of itself is the same as the MAC address of the new parent node or whether the MAC address of the new parent node is contained in a refugee table stored by itself, and then forwarding the third pairing cancellation packet downstream until the old parent node receives the third pairing cancellation packet; wherein if it determines that the MAC address of itself is the same as the MAC address of the new parent node or the MAC address of the new parent node is contained in the refugee table stored by itself, the relay node changes the sending address of the third pairing cancellation packet to the MAC address of itself, and then forwards the third pairing cancellation packet downstream; if it determines that the MAC address of itself is different from the MAC address of the new parent node or the MAC address of the new parent node is not contained in the refugee table stored by itself, the relay node deletes the MAC addresses of the network node and its descendant node from the refugee table stored by itself, changes the sending address of the third pairing cancellation packet to the MAC address of itself, and then forwards the third pairing cancellation packet downstream; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the third pairing cancellation packet.
15. The method according to claim 9, further comprising: generating and sending, by the root node corresponding to the new parent node, a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node to the old root node when the root node corresponding to the new parent node is different from a root node corresponding to the old parent node, wherein a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is a MAC address of the access point; deleting, by the old root node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream; deleting, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant node from a refugee table stored by itself, changing the sending

address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and deleting, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.

16. The method according to claim 9, further comprising: generating and sending, by any one relay node receiving the pairing change request packet or the root node corresponding to the new parent node, a pairing rejection packet when determining that it cannot provide relay services for the network node and its descendant node; wherein the pairing switching is failure when the network node receives the pairing rejection packet.

17. In a meshed network system for switching pairing, a node comprising a processor, memory, and software configured to: perform a scanning procedure to select a new parent node, so that a current parent node becomes an old parent node; generate and send a pairing change request packet carrying a MAC address of a descendant node of the network node, a MAC address of an old root node, and a MAC address of the old parent node to the new parent node when the new parent node is not an access point, wherein a sending address of the pairing change request packet is a MAC address of the network node, a receiving address of the pairing change request packet is a MAC address of the new parent node, and the old root node is a root node corresponding to the old parent node; generating and returning, by the new parent node receiving the pairing change request packet, a first pairing approval packet to the network node after determining that the MAC addresses of the network node and its descendant node are contained in a refugee table stored by itself, and generating and sending a second pairing cancellation packet carrying the MAC addresses of the network node and its descendant node downstream, wherein the refugee table stored by the new parent node contains MAC addresses of all subordinate nodes that the new parent node provides relay services for, a sending address of the first pairing approval packet is the MAC address of the new parent node, a receiving address of the first pairing approval packet is the MAC address of the network node, a receiving address of the second pairing cancellation packet is the MAC address of the old parent node, and a sending address of the second pairing cancellation packet is the MAC address of the new parent node; wherein pairing switching is successful if the network node receives the first pairing approval packet; delete, by a relay node receiving the second pairing cancellation packet, the MAC addresses of the network node and its descendant nodes from a refugee table stored by itself, changing the sending address of the second pairing cancellation packet to a MAC address of itself, and then forwarding the second pairing cancellation packet downstream until the old parent node receives the second pairing cancellation packet; and delete, by the old parent node, the MAC addresses of the network node and its descendant node from a refugee table stored by itself after receiving the second pairing cancellation packet.

18. The system according to claim 17, wherein the software is further configured to: generating and returning, by the new parent node, a first pairing approval packet to the network node when the new parent node is a root node, wherein a sending address of the first pairing approval packet is the MAC address of the new parent node and a receiving address of the first pairing approval packet is the MAC address of the network node; and forwarding, by the new parent node, the pairing change request packet upstream after changing the receiving address the pairing change request packet to a MAC address of its parent node when the new parent node is not the root node; wherein pairing switching is successful when the network node receives the first pairing approval packet.

19. The system according to claim 17, wherein the operation generating and sending, by the network node, the pairing change request packet carrying the MAC address of its descendant node, the MAC address of the old root node, and the MAC address of the old parent node to the new parent node further comprises: filling, by the network node, an address 1 field in a null data frame with the MAC address of the new parent node, filling an address 2 field in the null data frame with the MAC address of the network node, setting a first preset bit of an address 3 field in the null data

frame to a first preset value to define the null data frame as a handshake frame, setting a plurality of second preset bits of the address 3 field in the null data frame to a third preset value to define a type of the handshake frame as a pairing request, and filling an payload field in the null data frame, which is expanded, with the MAC address of the descendant node of the network node, the MAC address of the old root node, and the MAC address of the old parent node, so as to generate and send the pairing change request packet to the new parent node.

20. The system according to claim 18, wherein the operation of generating and returning, by the new parent node, the first pairing approval packet to the network node when the new parent node is the root node further comprises: filling, by the new parent node, an address 1 field in a null data frame with the MAC address of the network node, filling an address 2 field in the null data frame with the MAC address of the new parent node, setting a first preset bit of an address 3 field in the null data frame to a first preset value to define the null data frame as a handshake frame, and setting a plurality of second preset bits of the address 3 field in the null data frame to a fourth preset value to define a type of the handshake frame as pairing approval, so as to generate and return the first pairing approval packet.